SURJECTIVE MAPS IN HIGH DEGREES GIVE CLOSED IMMERSIONS OF PROJ

THE STACKS PROJECT AUTHORS

Editorial note. The counterexample closes with 'it is easy to see that r_psi^*O_X(1) is not zero' - a one-line argument (the pullback is invertible on the nonempty scheme Y); the surjectivity content is fully carried.

Allowed standard prerequisites (editorial).

- Morphisms from graded ring maps. lemma-morphism-proj (carded this batch). Supplies the characterized triple (U(psi), r_psi, theta).

Source reference. Original lemma 01MY: sources/stacks-constructions/pinned-original-source.tex, source lines 2271–2271. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *With hypotheses and notation as in Lemma 01MY above. Assume $A_d \to B_d$ is surjective for all $d \gg 0$. Then*

(1) *$U(\psi) = Y$,*
(2) *$r_\psi : Y \to X$ is a closed immersion, and*
(3) *the maps $\theta : r_\psi^* \mathcal{O}_X(n) \to \mathcal{O}_Y(n)$ are surjective but not isomorphisms in general (even if $A \to B$ is surjective).*

Proof. Part (1) follows from the definition of $U(\psi)$ and the fact that $D_+(f) = D_+(f^n)$ for any $n > 0$. For $f \in A_+$ homogeneous we see that $A_{(f)} \to B_{(\psi(f))}$ is surjective because any element of $B_{(\psi(f))}$ can be represented by a fraction $b/\psi(f)^n$ with n arbitrarily large (which forces the degree of $b \in B$ to be large). This proves (2). The same argument shows the map

$$A_f \to B_{\psi(f)}$$

is surjective which proves the surjectivity of θ. For an example where this map is not an isomorphism consider the graded ring $A = k[x, y]$ where k is a field and $\deg(x) = 1$, $\deg(y) = 2$. Set $I = (x)$, so that $B = k[y]$. Note that $\mathcal{O}_Y(1) = 0$ in this case. But it is easy to see that $r_\psi^* \mathcal{O}_X(1)$ is not zero. (There are less silly examples.) □

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.